Finding 20–60 second promos in a microdrama screenplay

Sandeep Kumar Jha · 9 November 2026

Kuku AI Generalist assignment · Problem 1 · 50 episodes, 102.7 minutes, 3,727 timestamped beats · Python 3, stdlib only

```
python3 -m promofinder --input ".." --out outputs --top 15  →  open outputs/report.html
```

1. FRAMING AND ASSUMPTIONS

The brief is precise about what a promo is, so I made its three tests the *gate*, not a score: a clip ships only if a hook lands inside the first three seconds, the cut leaves an unresolved moment, and a newcomer can follow it. Ranking is a separate, later question. Mixing the two is how a tool ends up confidently posting a bad clip.

- **Input is the screenplay pack only.** No video or audio, so every judgement is made on text and timecodes. I did not listen to any cut.
- **A promo may be one continuous stretch or two segments joined inside the same episode.** I did not join across episodes: the 50 files share characters but not a continuous timeline I can trust, and a cross-episode jump is a much harder standalone-sense problem.
- **The output is what an editor needs to make the cut,** not a rendered file: episode, in/out per segment, the hook line, the cliff line, and a one-line rationale. Subtitle and music choices are out of scope.
- **The show's own emotion parentheticals are free supervision.** Every line is tagged `(Devastated)`, `(Cold, Menacing)`; that turned out to be the single strongest hook feature, so the lexicon is built around it.

2. HOW IT WORKS

Parse every camera, action, sound and dialogue block into absolute-time beats; propose candidates; score them; rank. Boundaries are arithmetic, so rules do them exactly: 12 anchors per episode × 3 out-points, plus the episode's own ending, = **459 candidates (9.2/episode)**, all snapped to beat boundaries.

FEATURE	HOW IT IS MEASURED	WEIGHT
Hook	best beat in the first 3s: emotion intensity, short high-voltage line, reveal / slap / hard cut	0.30
Cliff	the *last spoken line*: question, trailing ellipsis, threat, reveal, plus a cut-to-black tag	0.30
Standalone	series-wide character debt: names the viewer has not met (e.g. "the Dowager"), alias-merged across all 50 files	0.22
Arc	emotional peak position and lift from opening to peak	0.12
Pacing	speech-to-runtime ratio and cuts per 10s	0.06

Score = 100 × weighted sum × duration fit (full credit 28–45s, small decay outside). A clip must also clear **0.45 hook / 0.45 cliff / 0.50 standalone**. Episodes that clear nothing are reported as "no postable promo" with the reason, not forced into the list. Two editorial passes run before the gate: **tail tightening** (cut at the last strong line if everything after it deflates) and **stitching** (join a strong opener to a strong closer, but only if the montage ends on a spoken line, adds ≥0.15 to hook+cliff, and loses ≤1.0 point elsewhere). Final ranking takes one pick per episode first, then a second pass, rejecting overlaps above 40%.

3. CODE, MODEL, HUMAN — WHAT DID WHICH JOB

CODE	MODEL	HUMAN (ME)	WHY
Parsing, timecodes, boundary snapping, length limits, gate arithmetic, dedupe, ranking, JSON/HTML. All deterministic; same pack → same list.	An optional judge over the top 2 clips per episode: hook / cliff / standalone / spoiler risk / postable, plus a reason and one fix. Blended 55% model, 45% rules. Never trusted on timestamps.	Chose the problem and the three-test framing; set the weights by looking at the score distribution; read all 15 top transcripts; overrode the tool's #1 pick; decided when to stop tuning.	Boundaries are arithmetic and should not cost a token. "Would a stranger understand this?" is judgement and should not be a regex. Tuning is neither.

The model is swappable by environment variable only — OpenAI-compatible (`PROMO_LLM_BASE_URL`, so Groq, OpenRouter, vLLM, Ollama all work) or Anthropic. No key ⇒ the heuristic pass runs and says so. Verdicts are cached by content hash.

4. QUALITY — WHAT THE RUN ACTUALLY PRODUCED

39/50	9/15	15/15	0	6.1s
EPISODES WITH A POSTABLE CUT	TOP PICKS SHIP AS-IS	SHIP AFTER ONE TRIM	UNUSABLE PICKS	WALL CLOCK, 50 EPISODES

I graded every top-15 pick by hand against the brief (`audit/quality_review.md`). Order: the tool ranks ep18 #1, but **I would post ep36 first** — its hook is spoken at 01:57.50 and lands in 1.8s, where ep18's hook is an image plus a question that needs the burning cottage to register first. That override is the human-in-the-loop working as intended.

#	EP	IN–OUT (S)	HOOK	CLIFF	VERDICT
1	18	145.8–186.8	"Why would anyone do this?"	"What are you going to do?"	Post (top 3)
2	36	117.5–148.3	"You told them the truth!"	"What wedding?"	**Post — first**
3	45	22.9–42.1 + 93.2–113.3	"You're the Duke?"	"You are not… lawful wife."	Post (stitched)
4	22	96.0–131.0	"Do not stare. Keep walking."	"Who has been treating you?"	Post
5	35	111.0–145.3	"How did you escape?"	"Who?"	Post
6	1	75.0–119.3	"…he was never going to marry you."	"…what I was worth to you?"	Post + title card

Three defects the audit found, all fixed and now covered by tests:

1. Reported cliff ≠ actual ending. The scorer took the highest-scoring line in the last 4s. On ep32 that was "A disgraced woman?", while the clip ran on to "…You cannot destroy me." — a resolved, triumphant ending. The card was describing a cut that was never made. The cliff is now the last spoken beat. Ep32 and ep3 dropped out (correctly); ep18's better narrative cut surfaced at #1.
2. Clips opened mid-sentence. The screenplays split lines across blocks ("…no one will remember…"). An ep29 pick started on the back half. The in-point now walks back to the sentence's first beat (max 10s).
3. Stitching won for the wrong reason. A flat score margin let ep1's montage win by jumping 35s and ending on "Cut to black", while ep45's genuinely good montage lost. Stitching now requires both bookends to improve, a spoken final line, and a bounded loss elsewhere.

Known bad cases I did not paper over. Three picks open on a fragment that answers an unheard question (#11 "No.", #12 "It is medicine!", #14 "I didn't!") — a reply-aware hook penalty would fix it but costs recall on punchy one-liners, so it is a human trim. Two picks have a procedural middle (#8 measuring a pulse, #9 reciting a journal) yet score pacing 1.00, because pacing only sees speech ratio and cut rate, not new information. ep50's best hook — "Absolution is free. Discretion has a price." — was *refused* at 0.42 because the line is delivered flat and has no emotion tag: the tags are doing a lot of the work. Of the 11 refusals, I judge 3 (ep21, ep39, ep49) to be false negatives and 6 clearly right; the rest fail on a visual hook the text cannot see.

5. SCALE, COST, AND THE NEXT HOUR OF WORK

At volume. Heuristic pass: ~0.12s per episode on one core, embarrassingly parallel. Model pass: 2 clips per episode, ~1,150 input / ~120 output tokens per call — about **$0.02 per 50-episode season** on gpt-4o-mini, $0.14 on Haiku, $0.50 on Sonnet at list prices; caching makes re-runs free. 500 episodes is ~1 minute of rules and 20 cents of models.

Human review is the real cost. Fifteen full transcripts took ~35 minutes. At 500 episodes, review only the top 40 and a random 10 for precision; everything else is a backlog. The tool is built to make that queue small and honest.

If I had a second day:

- A "reply-aware" hook penalty for one-word answers, and a `cliff_kind: visual` flag so wordless endings are phrased as an image, not blocking.
- An information-delta feature for pacing (new facts per 10s), so procedural scenes stop scoring 1.00.
- A series-trailer mode: 3–5 moments across episodes. The stitcher already builds the graph; the missing piece is a cross-episode standalone check.
- Test the 15 cuts against retention on a real account — no text heuristic replaces that signal.

6. THE POSTING PLAN I WOULD ACTUALLY RUN

1. **ep36 (31s), first.** Spoken accusation at 0.0s, four escalations, cliff on "What wedding?" — the cleanest test of a pure-dialogue cold open.
2. **ep22 (35s), second.** No names needed: sneaking, a dying woman, "Who has been treating you?" — the test of a premise-free visual open.
3. **ep45 (39s, stitched), third.** Reveal then reversal; add a 4-frame white flash at the join so the jump reads as a cut — the test of two-moment promos.
4. **ep18 (41s), fourth.** The tool's #1, held back on purpose: image-led hook, best ending.

Prototype: `promo-finder/` (stdlib only; the two commands above). Report: `outputs/report.html`, self-contained and offline. Raw run: `audit/run.log` · transcript sheet: `audit/top15_transcripts.txt` · hand review: `audit/quality_review.md` · 35 tests: `python3 -m unittest discover -s tests`.